

Scenario 02 — SmartGrid Meter (Energy)

Trainee handbook

🕒 Estimated time: **2 hours** · 🎯 Level: **Intermediate** · ⚡ Sector: **Energy (essential entity under NIS2)**

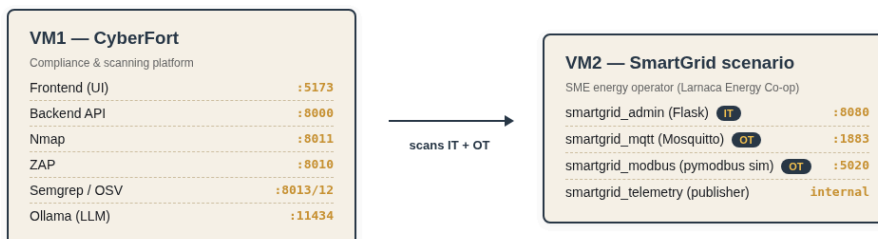
Scenario brief

You are the new cybersecurity lead at **Larnaca Energy Co-op**, an SME distribution-system operator. The co-op runs a small fleet of smart meters across Larnaca, Paphos and Nicosia. The meters publish telemetry to an in-house **MQTT** broker every few seconds and the operations team reads register values directly from each meter over **Modbus TCP** through a small web console called **SmartGrid Meter Admin v1.3.0**.

Because the Co-op qualifies as an **essential entity** under Directive (EU) 2022/2555 (NIS2), the board wants a full security review of the meter fleet's management plane before the next supervisory inspection. You will use **CyberFort** to scan, score, and produce a NIS2 readiness report.

You have:

- A virtual machine running CyberFort (`https://<VM1-IP>:5173`), or access to the hosted reference instance at `https://access.cyber-fort.eu/login`.
- A virtual machine hosting this scenario in Docker (`<VM2-IP>`). Shell access to VM2.
- CyberFort credentials issued by your instructor.



i The hosted reference instance cannot reach a private cyber-range subnet, so the scanner integrations cannot actively scan VM2 from there. The walkthrough below shows the **exact UI steps** for each scan — and where the scanner cannot reach your target, you **replicate** the finding by adding it manually to the Risk Register.

Learning outcomes

By the end of this scenario you will be able to:

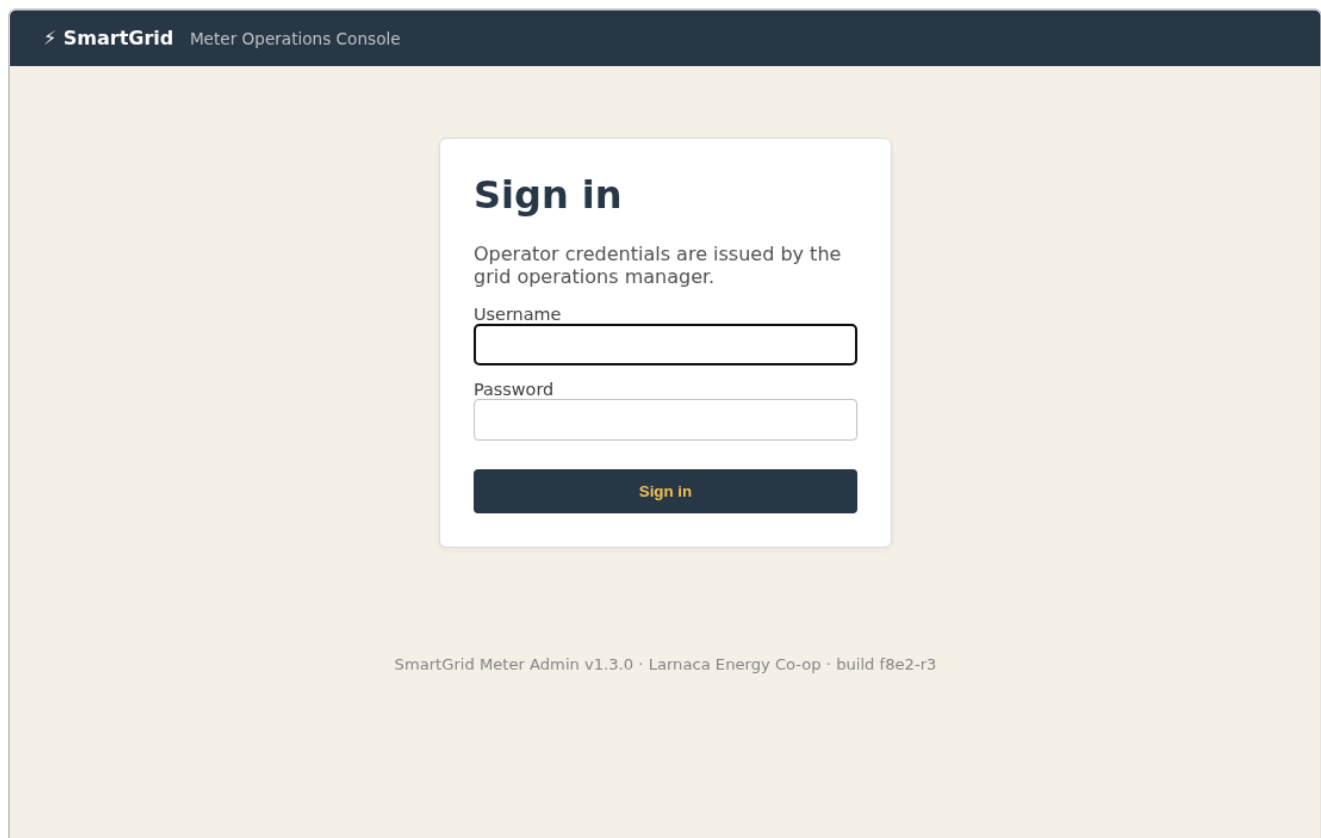
1. Register a multi-component product (IT + OT) in CyberFort.
2. Run an **Nmap** scan covering both IT and **OT** ports (8080, 1883, 5020).
3. Verify anonymous **MQTT** subscription and unauthenticated **Modbus** register reads.
4. Run an **OWASP ZAP** scan against the management web UI and uncover default credentials + a firmware-upload RCE.
5. Run a **Semgrep** SAST scan and find hardcoded credentials in source code.
6. Run an **OSV** dependency scan against `requirements.txt`.
7. Complete a **NIS2 Article 21(2)** assessment in CyberFort.
8. Export a NIS2 readiness PDF report.

Step 0 — Verify the lab is up

On VM2:

```
cd /srv/smartgrid-scenario
docker compose ps
```

You should see four running containers: `smartgrid_admin`, `smartgrid_mqtt`, `smartgrid_modbus`, `smartgrid_telemetry`. Open `http://<VM2-IP>:8080/login` in a browser. You should land on the SmartGrid Meter Admin sign-in page:



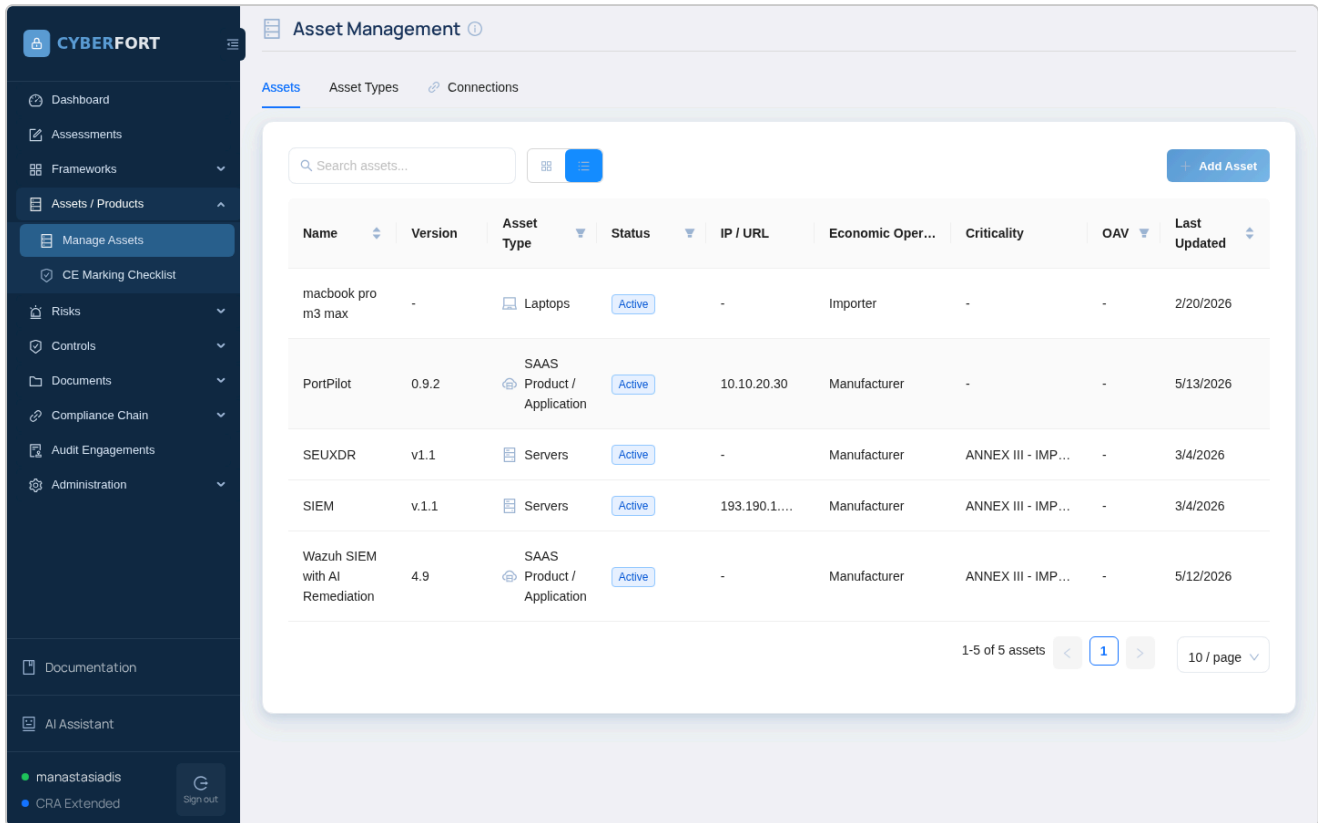
The image shows a web browser window displaying the SmartGrid Meter Admin sign-in page. The page has a dark blue header with the SmartGrid logo and the text 'Meter Operations Console'. The main content area is light beige. In the center, there is a white box with the title 'Sign in'. Below the title, it says 'Operator credentials are issued by the grid operations manager.' There are two input fields: 'Username' and 'Password'. Below these fields is a dark blue button with the text 'Sign in' in yellow. At the bottom of the page, there is a footer that reads 'SmartGrid Meter Admin v1.3.0 · Larnaca Energy Co-op · build f8e2-r3'.



Do **not** try to sign in yet. We will let CyberFort do the discovery for us.

Step 1 — Register the meter management system as a product in CyberFort

1. From the left sidebar, expand **Assets / Products** and click **Manage Assets**. You see the existing asset list.

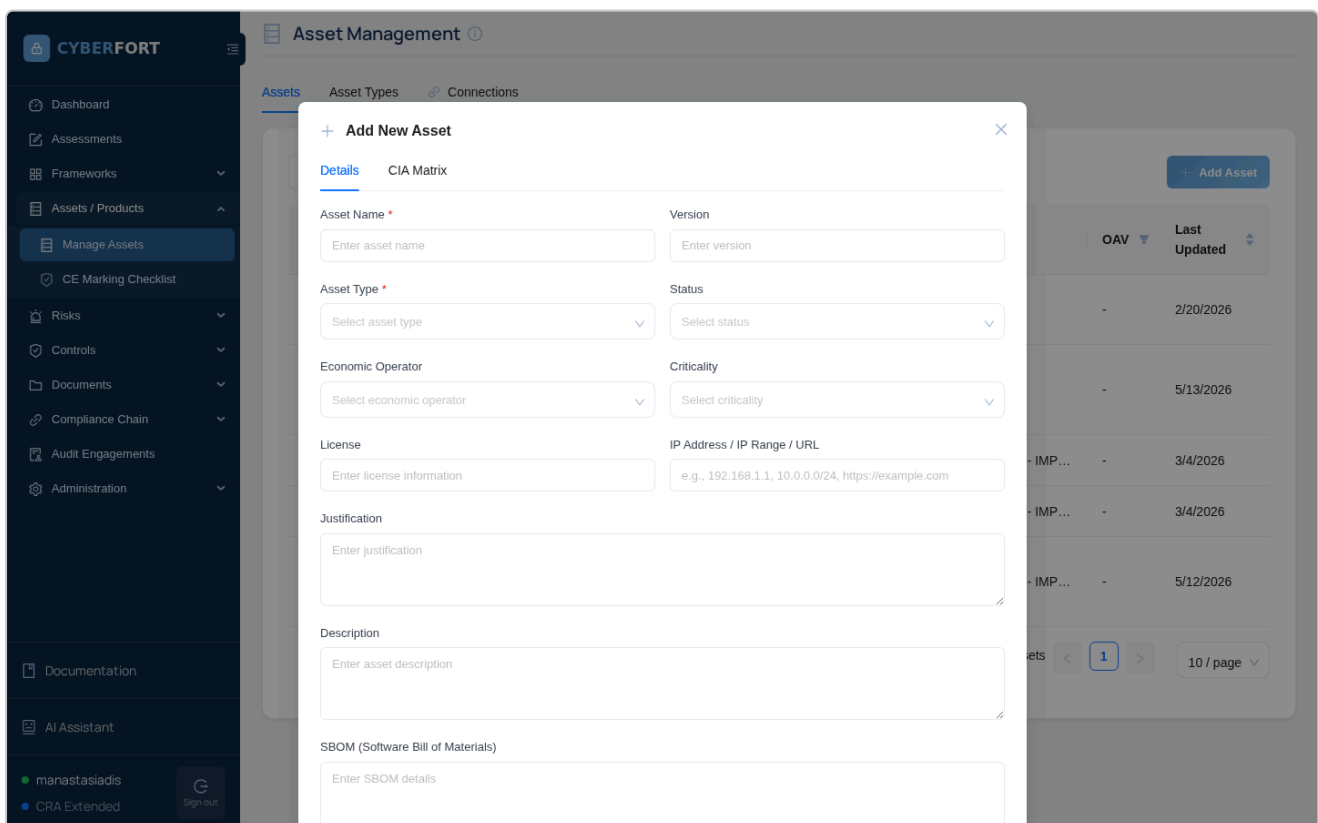


The screenshot shows the CyberFort Asset Management interface. The left sidebar contains the following menu items: Dashboard, Assessments, Frameworks, Assets / Products (expanded), Manage Assets (selected), CE Marking Checklist, Risks, Controls, Documents, Compliance Chain, Audit Engagements, and Administration. The main content area is titled "Asset Management" and includes tabs for Assets, Asset Types, and Connections. A search bar is present at the top of the asset list. The asset list table has the following columns: Name, Version, Asset Type, Status, IP / URL, Economic Oper..., Criticality, OAV, and Last Updated. The table contains five rows of assets, all with a status of "Active".

Name	Version	Asset Type	Status	IP / URL	Economic Oper...	Criticality	OAV	Last Updated
macbook pro m3 max	-	Laptops	Active	-	Importer	-	-	2/20/2026
PortPilot	0.9.2	SAAS Product / Application	Active	10.10.20.30	Manufacturer	-	-	5/13/2026
SEUXDR	v1.1	Servers	Active	-	Manufacturer	ANNEX III - IMP...	-	3/4/2026
SIEM	v.1.1	Servers	Active	193.190.1...	Manufacturer	ANNEX III - IMP...	-	3/4/2026
Wazuh SIEM with AI Remediation	4.9	SAAS Product / Application	Active	-	Manufacturer	ANNEX III - IMP...	-	5/12/2026

At the bottom right of the table, there is a pagination control showing "1-5 of 5 assets" and a page number "1" out of "10 / page".

2. Click **+ Add Asset** (top right). The *Details* tab of the modal opens.



The screenshot shows the CyberFort Asset Management interface with the "Add New Asset" modal open. The modal has two tabs: "Details" (selected) and "CIA Matrix". The "Details" tab contains the following fields:

- Asset Name * (text input)
- Version (text input)
- Asset Type * (dropdown menu)
- Status (dropdown menu)
- Economic Operator (dropdown menu)
- Criticality (dropdown menu)
- License (text input)
- IP Address / IP Range / URL (text input, example: e.g., 192.168.1.1, 10.0.0.0/24, https://example.com)
- Justification (text input)
- Description (text input)
- SBOM (Software Bill of Materials) (text input)

The modal is overlaid on the Asset Management interface, which is dimmed in the background.

3. Fill in: * **Asset Name:** SmartGrid Meter Admin * **Version:** 1.3.0 * **Asset Type:** SAAS Product / Application * **Status:** Active * **Economic Operator:** Manufacturer * **Criticality:** ANNEX III (the meter fleet supports an essential service) * **IP Address / URL:** <VM2-IP> (e.g. 10.10.20.31) * **Description:** "SmartGrid Meter Admin — Flask portal for an SME energy operator. Manages a fleet of 3 smart meters via Modbus TCP and MQTT."

The screenshot shows the 'Add New Asset' form in the CYBERFORT Asset Management interface. The form is titled 'Add New Asset' and has a 'Details' tab selected. The form fields are as follows:

- Asset Name:** SmartGrid Meter Admin
- Version:** 1.3.0
- Asset Type:** SAAS Product / Application
- Status:** Active
- Economic Operator:** Manufacturer
- Criticality:** Select criticality
- License:** Enter license information
- IP Address / IP Range / URL:** 10.10.20.31
- Justification:** Enter justification
- Description:** SmartGrid Meter Admin — Flask portal for an SME energy operator (Larnaca Energy Co-op). Manages a fleet of 3 smart meters via Modbus TCP and MQTT. Trains operators on the SmartGrid cyber-range scenario.
- SBOM (Software Bill of Materials):** Enter SBOM details

The background shows a list of assets with columns: Name, Version, Asset Type, Status, IP / URL, Economic Oper..., Criticality, OAV, and Last Updated. The 'Add Asset' button is visible in the top right corner.

4. Click **Save**. SmartGrid Meter Admin now appears in the asset list.

The screenshot shows the 'Asset Management' interface with the 'Assets' tab selected. The asset list is displayed with the following columns: Name, Version, Asset Type, Status, IP / URL, Economic Oper..., Criticality, OAV, and Last Updated. The 'SmartGrid Meter Admin' asset is now visible in the list.

Name	Version	Asset Type	Status	IP / URL	Economic Oper...	Criticality	OAV	Last Updated
macbook pro m3 max	-	Laptops	Active	-	Importer	-	-	2/20/2026
PortPilot	0.9.2	SAAS Product / Application	Active	10.10.20.30	Manufacturer	-	-	5/13/2026
SEUXDR	v1.1	Servers	Active	-	Manufacturer	ANNEX III - IMP...	-	3/4/2026
SIEM	v1.1	Servers	Active	193.190.1...	Manufacturer	ANNEX III - IMP...	-	3/4/2026
SmartGrid Meter Admin	1.3.0	SAAS Product / Application	Active	10.10.20.31	Manufacturer	-	-	5/13/2026
Wazuh SIEM with AI Remediation	4.9	SAAS Product / Application	Active	-	Manufacturer	ANNEX III - IMP...	-	5/12/2026

The background shows the 'Add Asset' button in the top right corner. The bottom of the page shows pagination: 1-6 of 6 assets, 1 / page.

Step 2 — Run an Nmap scan against the OT/IT stack

The critical step. Energy operators must know **every exposed port** on the management plane.

1. Expand **Security Tools** in the sidebar and click **Security Scanners**.
2. Scan Type: **Basic Scan - Top 1000 ports** . Target: **<VM2-IP>** .
3. Click **Run Scan**.

The screenshot displays the CYBERFORT Security Scanners interface. The left sidebar contains navigation links: Dashboard, Assessments, Frameworks, Assets / Products, Risks, Controls, Documents, Compliance Chain, Audit Engagements, and Administration. The main content area is titled 'Security Scanners' and includes a 'Scanner Guide' button. Below the title, there are tabs for 'Network Vulnerability' (selected) and 'Application Vulnerability'.

The 'Scan Configuration' section shows the 'Scan Type' set to 'Basic Scan - Top 1000 ports' and the 'Target' field with the placeholder 'IP address, domain, or network range'. There are radio buttons for 'Enter Custom IP/Domain' (selected) and 'Select from Assets'. At the bottom of this section are buttons for 'Run Scan', 'Clear', and 'Schedule'.

The 'Scan Results' section shows a list of 12 scans. The table below contains the data for these scans.

	Timestamp	Scan Type	Status	Severity	Duration (s)	User	Organization	Acti
>	Apr 25, 2026, 09:07:48 AM	basic	COMPLETED	INFO	39.27	nkekatos@clone-system...	Clone Systems	
>	Apr 25, 2026, 09:06:38 AM	aggressive	COMPLETED	INFO	0.67	nkekatos@clone-system...	Clone Systems	
>	Apr 25, 2026, 09:06:02 AM	basic	COMPLETED	INFO	0.45	nkekatos@clone-system...	Clone Systems	
>	Mar 14, 2026, 11:29:25 AM	basic	COMPLETED	INFO	20.65	nkekatos@clone-system...	Clone Systems	
>	Feb 26, 2026, 01:56:22 PM	fast	COMPLETED	INFO	19.08	superadmin@clone-syst...	Clone Systems	
>	Feb 26, 2026, 01:43:07 PM	basic	COMPLETED	INFO	23.15	superadmin@clone-syst...	Clone Systems	
>	Jan 28, 2026, 08:06:20 AM	os	COMPLETED	N/A	13.73	superadmin@clone-syst...	Clone Systems	
>	Nov 24, 2025, 09:36:04 AM	basic	COMPLETED	N/A	16.22	nmiaritis@clone-system...	Clone Systems	
>	Nov 21, 2025, 07:57:09 PM	all_ports	COMPLETED	N/A	225.05	superadmin@clone-syst...	Clone Systems	
>	Nov 21, 2025, 08:25:52 AM	basic	COMPLETED	N/A	5.47	nmiaritis@clone-system...	Clone Systems	

At the bottom of the table, there is a pagination control showing '1-10 of 12 scans', a page number '1' in a blue box, a '2' button, and a '10 / page' dropdown. There are also 'Go to' and 'Page' input fields.

After ~1 minute you should see:

Port	Service	Notes
8080	http	SmartGrid Meter Admin web UI
1883	mqtt	MQTT broker — should never be exposed outside the OT subnet
5020	modbus-tcp	Modbus TCP — protocol has no authentication by design

🔴 **Finding #1 — MQTT broker exposed.** MQTT on 1883 is in clear text and (as we are about to discover) accepts anonymous clients. NIS2 Article 21(2)(d), (e), (h).

🔴 **Finding #2 — Modbus TCP exposed.** Modbus has no built-in authentication; anyone who reaches port 5020 can read or write meter registers. NIS2 Article 21(2)(b), (e).

Step 3 — Confirm the exposures manually

CyberFort's scanners flag the ports, but you need concrete evidence to file in the Risk Register.

3a. MQTT — anonymous subscribe

```
mosquitto_sub -h <VM2-IP> -p 1883 -t 'smartgrid/#' -v -W 8
```

Expected output:

```
smartgrid/meter-larnaca-001/telemetry {"meter_id": "meter-larnaca-001", "ts": ..., "power_w": 5299, "v
smartgrid/meter-paphos-014/telemetry {"meter_id": "meter-paphos-014", ...}
smartgrid/meter-nicosia-027/telemetry {"meter_id": "meter-nicosia-027", ...}
```

Three meters' real-time kWh and tariff data, in clear text, with no authentication.

🔴 **Finding #3 — Anonymous MQTT subscription leaks consumption data.** NIS2 Article 21(2)(h) — *use of cryptography*. Customer kWh is personal data with a GDPR dimension.

3b. Modbus — unauthenticated register read

```
python -c "
from pymodbus.client.sync import ModbusTcpClient
c = ModbusTcpClient('<VM2-IP>', port=5020); c.connect()
print(c.read_holding_registers(0, 9, unit=1).registers)
"
```

Expected output:

```
[4121, 2282, 2301, 1116, 1048, 1350, 2, 6346, 324]
# power_w · voltage L1×10 · voltage L2×10 · voltage L3×10 · current L1×100 · current L2×100 · current
```

A meter could be written to in the same way (function code 6 — write single register).

 **Finding #4 — Modbus registers readable and writable by anyone on the network.** NIS2 Article 21(2)(b), (e), (i).

The same data is visible through the management UI (which uses the same protocol internally):

 Meter Operations Console

DashboardFirmwareadminLogout

Meter #1 — Modbus holding registers

Live read from the meter over Modbus TCP.

#	Description	Raw value
---	-------------	-----------

[« Back to dashboard](#)

SmartGrid Meter Admin v1.3.0 · Larnaca Energy Co-op · build f8e2-r3

Step 4 — Run an OWASP ZAP scan against the management web UI

1. **Security Tools** → **Security Scanners** → **Application Vulnerability** tab.
2. Target: `http://<VM2-IP>:8080` . Active scan.
3. While ZAP runs, try the default credentials yourself: `admin` / `admin` . They work.

SmartGrid

Meter Operations Console

Sign in

Operator credentials are issued by the grid operations manager.

Username

admin

Password

.....

Sign in

SmartGrid Meter Admin v1.3.0 · Larnaca Energy Co-op · build f8e2-r3

After signing in you land on the fleet dashboard:

SmartGrid

Meter Operations Console

DashboardFirmwareadminLogout

Fleet telemetry

Latest MQTT samples from each meter. Click a meter to read its Modbus registers.

Meter	Power (W)	Voltage (V)	Total kWh	Tariff	Registers
meter-larnaca-001	2861	229.52	138393.0	day	Read Modbus #1
meter-paphos-014	3685	229.08	127087.1	night	Read Modbus #2
meter-nicosia-027	3031	230.47	128351.9	night	Read Modbus #3

SmartGrid Meter Admin v1.3.0 · Larnaca Energy Co-op · build f8e2-r3

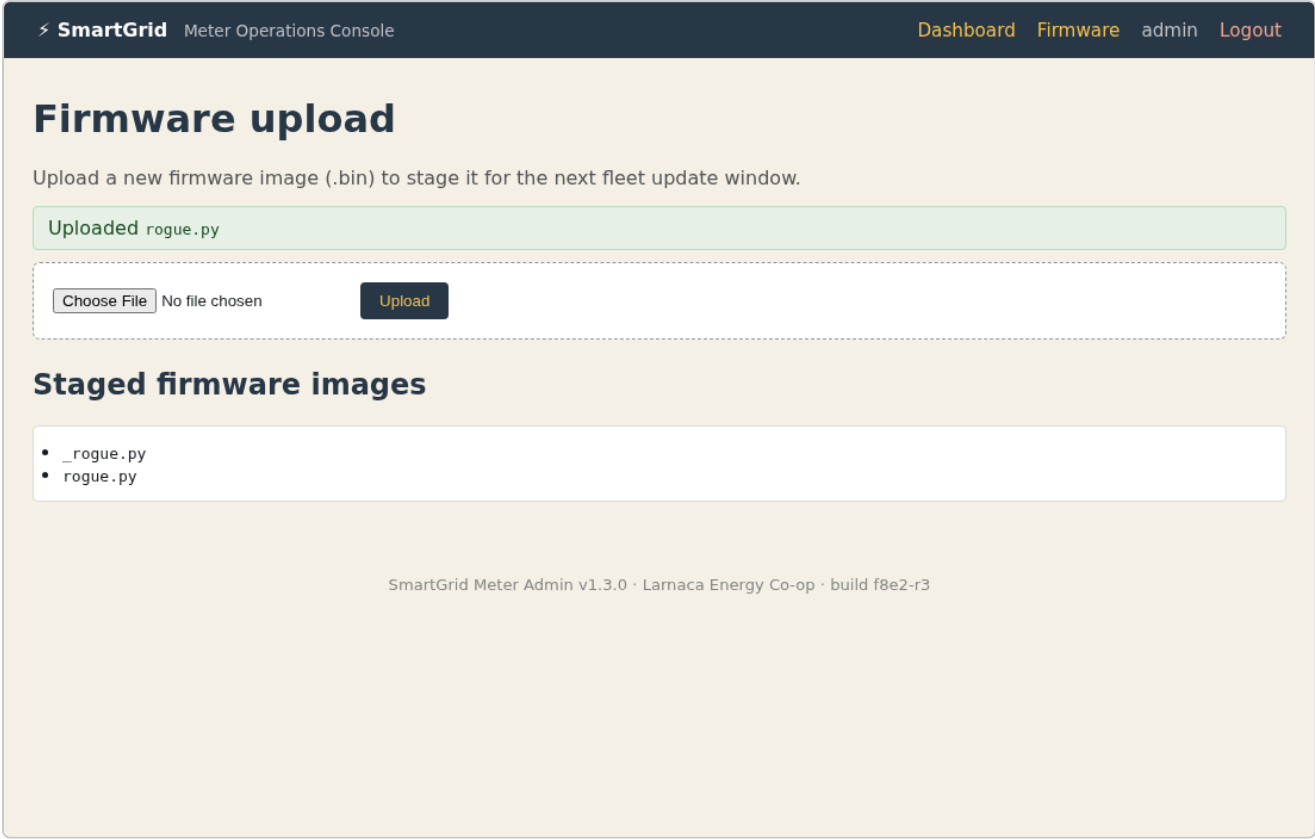
Expected ZAP findings:

- **Default Credentials** — admin/admin on /login . Severity High.
- **File Upload — No Type Restriction** on /firmware . Severity High.
- **CSRF** on /firmware (no token).

- **Cookie flags missing** — session cookie without `Secure` or `HttpOnly`.

4a. Firmware upload accepts arbitrary file types

The `/firmware` endpoint stages any file under the meter-admin uploads directory — try uploading a `.py` file:



- **Finding #5 — Default operator credentials shipped with the product.** NIS2 Article 21(2)(i).
- **Finding #6 — Arbitrary file upload on the firmware endpoint.** NIS2 Article 21(2)(d), (e) — supply-chain & firmware integrity.

Step 5 — Run a Semgrep SAST scan

```
cd /srv/smartgrid-scenario
zip -r /tmp/smartgrid-src.zip admin/ modbus/ telemetry/
```

In CyberFort: **Security Tools** → **Code Analysis**. Upload the ZIP and **Run Scan**.

- Dashboard
- Assessments
- Frameworks
- Assets / Products
- Risks
- Controls
- Documents
- Compliance Chain
- Audit Engagements
- Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Sign out

Code Analysis

Scanner Guide

Scan Configuration

Configuration *

auto

Analysis Mode

LLM Analysis

Fast Results

Source Code

Upload ZIP File

GitHub Repository

Click to select a ZIP file

Upload a ZIP file containing source code for security analysis

Run Scan

Clear

Schedule

Scan Results 99+

Search history...

	Timestamp	Scan Type	Status	Severity	Duration (s)	User	Organization	Acti
>	May 13, 2026, 09:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 08:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 07:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 06:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 05:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 04:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 03:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 02:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 01:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 12:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	

1-10 of 100 scans

1

2

3

4

5

...

10

10 / page

Go to

Page

Expected findings:

Severity	Rule	File
High	hardcoded-password	telemetry/publisher.py (MQTT_PASSWORD = "OpsTopSecret#2024")

Severity	Rule	File
High	hardcoded-password	admin/meter_admin/config.py (BILLING_API_KEY)
Medium	flask-debug-true	admin/meter_admin/app.py
Medium	arbitrary-file-write	admin/meter_admin/app.py (firmware upload)

 **Finding #7 — Hardcoded credentials in source.** NIS2 Article 21(2)(j) — no operational secret should live in source.

Step 6 — Run an OSV dependency scan

Security Tools → **Dependency Check**. Upload the same ZIP or just `admin/requirements.txt`.

- Dashboard
- Assessments
- Frameworks
- Assets / Products
- Risks
- Controls
- Documents
- Compliance Chain
- Audit Engagements
- Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Dependency Analysis

Scanner Guide

Source Code

Upload ZIP File

GitHub Repository

Click to select a ZIP file

Upload a ZIP file containing source code to scan for dependency vulnerabilities

Scan Control

LLM Analysis

Fast Results

Run Scan

Clear

Schedule

Scan Results

99+

Search history...

Grid

Table

	Timestamp	Scan Type	Status	Severity	Duration (s)	User	Organization	Actions
>	May 13, 2026, 09:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 09:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 08:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 08:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 07:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 07:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 06:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 06:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 05:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	
>	May 13, 2026, 05:59:07 AM	basic	FAILED	N/A	0.00	manastasiadis@clone-s...	Clone Systems	

1-10 of 100 scans

1

2

3

4

5

...

10

10 / page

Go to

Page

Expected advisories:

Package	Version	Notable advisory
Flask	2.0.1	GHSA-m2qf-hxjv-5gpq

Package	Version	Notable advisory
Werkzeug	2.0.1	GHSA-px8h-6qyv-m22q, GHSA-2g68-c3qc-8985
Jinja2	3.0.0	GHSA-h5c8-rqwp-cp95
requests	2.25.0	GHSA-j8r2-6x86-q33q
pymodbus	2.5.3	GHSA-7796-x727-6w2v



Finding #8 — Multiple high-severity advisories on third-party components. NIS2 Article 21(2) (d).

Cross-check with Scan Findings

Security Tools → **Scan Findings** aggregates everything into one filterable table:

CYBERFORT

Dashboard
Assessments
Frameworks
Assets / Products
Risks
Controls
Documents
Compliance Chain
Audit Engagements
Administration

Documentation
AI Assistant
manastasiadis
CRA Extended
Sign out

Scan Findings

Total Findings: 316
Remediated: 0 / 316 (0%)
Linked to Risks: 31 / 316 (10%)

Dependency Scanner: 133
Network Scanner: 16
Code Scanner: 63
Web App Scanner: 104

MEDIUM: 181
HIGH: 12
INFO: 123

Scanner Type: [v] Severity: [v] Risk Link: [v] Remediated: [v] Search title, identi... [v] Reset Refresh

Scanner	Target	Findings	Date
Web App Scanner	https://www.draxis.gr/	11	4/25/2026, 9:10:24 AM
Web App Scanner	https://www.draxis.gr/	11	4/25/2026, 9:10:18 AM
Web App Scanner	https://www.draxis.gr/	11	4/25/2026, 9:10:10 AM
Network Scanner	www.draxis.gr/	5	4/25/2026, 9:07:48 AM
Network Scanner	www.draxis.gr/	1	4/25/2026, 9:06:38 AM
Network Scanner	www.draxis.gr/	1	4/25/2026, 9:06:02 AM
Web App Scanner	https://ktelachaias.gr/	10	3/14/2026, 12:39:01 PM

1-50 of 316 findings < 1 2 3 4 5 6 7 >

After filing the eight risks for SmartGrid Meter Admin, your **Risk Register** should look similar to this:

CYBERFORT

Dashboard
Assessments
Frameworks
Assets / Products
Risks
Risk Register
Risk Assessment
Incident Registration
Security Advisories
Controls
Documents
Compliance Chain
Audit Engagements
Administration
Documentation
AI Assistant
manastasiadis
CRA Extended
Sign out

Risk Management

Dashboard
Risk Registry 89
Risk Templates Library 110
Connections

Search risks...

Add New Risk
Export to PDF (89)

Code	Risk Category	Description	Status (Treatment)	Severity	Assessment	Likelihood
RSK-1	DNS Attacks	Attacks targeting Domain Name ...	Accept	Medium	Not Assessed	Medium
RSK-13	Phishing Attacks	Social engineering attacks that tr...	Accept	Medium	Assessment in progress	Low
RSK-30	Accidental fire, water,...	Risk of outages and loss due to ...	Accept	Medium	-	Medium
RSK-31	Availability & Disaste...	Risk of system outage due to a d...	Accept	Medium	-	Medium
RSK-32	Broken or weak auth...	Risk of account hijacking or expl...	Accept	Medium	-	Medium
RSK-33	Brute force attack	Risk of malicious user getting ac...	Accept	Medium	-	Medium
RSK-34	Cancelled contract	A contract is cancelled due to a ...	Accept	Medium	-	Medium
RSK-35	Cloud misconfiguration	Lack of baseline configuration wi...	Accept	Medium	-	Medium
RSK-36	DNS Spoofing	Risk of data theft and malware di...	Accept	Medium	-	Medium
RSK-37	Data breach	Open ports, unrestricted protocol...	Accept	Medium	-	Medium

1-10 of 89 risks
1 2 3 4 5 ... 9
10 / page

1. **Assessments** → + New Assessment.

CYBERFORT

Dashboard

Assessments

Frameworks

Assets / Products

Risks

Controls

Documents

Compliance Chain

Security Tools

Audit Engagements

Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Sign out

Assessments

Export PDFExport CSVImport CSVSuggest from ScansAI Suggest Answers

Assessment Overview

View your active assessments or create new ones

Active Assessments

Assessments currently in progress

No active assessments

Create New Assessment

Start a new assessment for a framework

+ New Assessment

0

Total

0

Completed

Select Assessment

Or browse and select from existing assessments

Framework

Assessment Type

Assessment

CRA

Select type

Select assessment

2. Fill the form: * **Framework:** NIS2 * **Assessment Type:** Conformity * **Scope Type:** Asset / Product * **Asset / Product:** SmartGrid Meter Admin * **Assessment Name:** SmartGrid NIS2 Conformity

CYBERFORT

Dashboard

Assessments

Frameworks

Assets / Products

Risks

Controls

Documents

Compliance Chain

Audit Engagements

Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Sign out

Assessments

Export PDF

Export CSV

Import CSV

Suggest from Scans

AI Suggest Answers

Assessment Overview

View your active assessments or create new ones

Active Assessments

2

Assessments currently in progress

PortPilot CRA Conformity

CRA • Product: PortPilot

0%

pentest

PCI DSS v4.0 • Organization: Clone Systems

0%

Create New Assessment

Start a new assessment for a framework

Framework

CRA

Assessment Type

Conformity

Scope Type *

Asset / Product

Asset / Product *

SmartGrid Meter Admin v1.3.0 (SAAS Product / Application)

Assessment Name

SmartGrid NIS2 Conformity

Create

Cancel

2

Total

0

Completed

Select Assessment

Or browse and select from existing assessments

Framework

CRA

Assessment Type

Conformity

Assessment

Select assessment

3. Click **Create**.

CYBERFORT

Assessments

Dashboard

Assessments

Frameworks

Assets / Products

Risks

Controls

Documents

Compliance Chain

Audit Engagements

Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Sign out

Assessments

Export PDF

Export CSV

Import CSV

Suggest from Scans

AI Suggest Answers

Assessment Overview

View your active assessments or create new ones

Active Assessments

2

Assessments currently in progress

PortPilot CRA Conformity

CRA • Product: PortPilot

0%

pentest

PCI DSS v4.0 • Organization: Clone Systems

0%

Create New Assessment

Start a new assessment for a framework

Framework

CRA

Assessment Type

Conformity

Scope Type *

Asset / Product

Asset / Product *

SmartGrid Meter Admin v1.3.0 (SAAS Product / Application)

Assessment Name

SmartGrid NIS2 Conformity

Create

Cancel

2

Total

0

Completed

Select Assessment

Or browse and select from existing assessments

Framework

CRA

Assessment Type

Conformity

Assessment

Select assessment

4. Click the new **SmartGrid NIS2 Conformity** card to open the 66-question questionnaire.

CYBERFORT

Dashboard

Assessments

Frameworks

Assets / Products

Risks

Controls

Documents

Compliance Chain

Audit Engagements

Administration

Documentation

AI Assistant

manastasiadis

CRA Extended

Sign out

Assessments

Export PDFExport CSVImport CSV

Suggest from ScansAI Suggest Answers

Assessment Overview

View your active assessments or create new ones

Active Assessments3

Assessments currently in progress

SmartGrid NIS2 Conformity

CRA • Product: SmartGrid Meter Admin

0%

PortPilot CRA Conformity

CRA • Product: PortPilot

0%

pentest

Create New Assessment

Start a new assessment for a framework

Framework

CRA

Assessment Type

Conformity

Scope Type

Select scope type

Assessment Name

Enter assessment name

Create

Cancel

3

Total

0

Completed

Select Assessment

Or browse and select from existing assessments

Framework

CRA

Assessment Type

Conformity

Assessment

SmartGrid NIS2 Conformity

Delete Assessment

Assessment Questions

(Asset / Product: SmartGrid Meter Admin)

Question 1

Mandatory: YESCompleted: NOCommon in: CRA(clone-systems.com)

Have you undertaken an assessment of the cybersecurity risks associated with your product with digital elements?

Yes

No

Partially

N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 2

Mandatory: YESCompleted: NOCommon in: CRA(clone-systems.com)

Do you take the outcome of the cybersecurity risk assessment into account during the planning, design, development, production, delivery and maintenance phases?

Yes

No

Partially

N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 3

Mandatory: YESCompleted: NOCommon in: CRA(clone-systems.com)

Is your cybersecurity risk assessment documented and updated as appropriate during the support period?

Save Answer

Clear Answer

Attach File(s)

☐ Yes ☐ No ☐ Partially ☐ N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Question 4

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Does your cybersecurity risk assessment comprise an analysis based on the intended purpose and reasonably foreseeable use of the product?

☐ Yes ☐ No ☐ Partially ☐ N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 5

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Does your risk assessment take into account the conditions of use, operational environment, and assets to be protected?

☐ Yes ☐ No ☐ Partially ☐ N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 6

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Does your risk assessment consider the length of time the product is expected to be in use?

☐ Yes ☐ No ☐ Partially ☐ N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 7

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Does your cybersecurity risk assessment indicate whether and how the security requirements are applicable to your product?

☐ Yes ☐ No ☐ Partially ☐ N/A

Assign Policy

No Policy

Evidence Description

Describe the evidence supporting this answer...

0 file(s) attached:

Save Answer

Clear Answer

Attach File(s)

Question 8

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Do you include the cybersecurity risk assessment in the technical documentation when placing the product on the market?

☐ Yes ☐ No ☐ Partially ☐ N/A

Save Answer

Clear Answer

Attach File(s)

Assign Policy
No Policy

Evidence Description
Describe the evidence supporting this answer...

0 file(s) attached:

Question 9

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Save Answer

Clear Answer

Attach File(s)

Where certain essential cybersecurity requirements are not applicable, do you include clear justification in the technical documentation?

☐ Yes
☐ No
☐ Partially
☐ N/A

Assign Policy
No Policy

Evidence Description
Describe the evidence supporting this answer...

0 file(s) attached:

Question 10

Mandatory: **YES** Completed: **NO** Common in: CRA(clone-systems.com)

Save Answer

Clear Answer

Attach File(s)

Do you systematically document relevant cybersecurity aspects concerning your products with digital elements?

☐ Yes
☐ No
☐ Partially
☐ N/A

Assign Policy
No Policy

Evidence Description
Describe the evidence supporting this answer...

0 file(s) attached:

1-10 of 52 questions
1 2 3 4 5 6

5. Focus on the questions below — they line up directly with your findings. **The text is exactly as you will see it in the CyberFort UI** (taken from the platform's NIS2 question pool).

#	Question (verbatim from the CyberFort NIS2 questionnaire)	Your answer	Evidence
1	"Are cybersecurity risk analysis policies documented and do they address sector-specific cybersecurity risks?"	Not compliant	No documented OT risk policy.
3	"Are incident handling procedures documented covering detection, response, and recovery phases of cybersecurity incidents?"	Not compliant	No SIEM, no auth-failure logs, no playbook for an MQTT compromise.
7	"Are supply chain cybersecurity risks assessed and managed through formal procedures and contractual requirements?"	Not compliant	OSV finding — five outstanding advisories on shipped libraries.
9	"Are security requirements integrated into system acquisition, development, and maintenance processes, including vulnerability management?"	Not compliant	Hardcoded secrets + debug mode + default credentials.

#	Question (verbatim from the CyberFort NIS2 questionnaire)	Your answer	Evidence
11	"Are policies and procedures established to regularly assess and measure cybersecurity risk management measure effectiveness?"	Not compliant	First effectiveness review is this exercise.
13	"Are basic cyber hygiene practices implemented and is cybersecurity training provided to relevant personnel?"	Partially	Operators received generic IT training but no OT-specific module.
15	"Are policies and procedures for cryptography and encryption use documented and implemented based on risk assessment?"	Not compliant	MQTT in clear text; no TLS on the broker.
17	"Are human resources security measures, access control policies, and asset management procedures documented and implemented?"	Not compliant	Default admin/admin ; one role for every operator; no asset-management of meter inventory.
19	"Are multi-factor authentication and secure communication systems implemented where appropriate, particularly for cybersecurity incident management personnel?"	Not compliant	No MFA on the management portal.
53	"Do cybersecurity measures address the critical infrastructure nature and dependencies of essential entity operations?"	Not compliant	Modbus + MQTT exposed; no segmentation between management plane and customer-data telemetry.

6. Attach the relevant scan output as evidence on each "Not compliant" answer.

7. Save as **Draft**.

 Use the **AI assistant** on at least two questions — for example, ask it to draft a remediation paragraph for the MQTT clear-text finding.

Step 8 — Generate the NIS2 readiness PDF

Use the **Export PDF** button on the Assessments page. Combine it with the **Export to PDF** on the Risk Register dashboard for a complete readiness pack — the document an SME would hand to its NIS2 supervisory authority.

Step 9 — Stretch goals

- Patch one vulnerability (`mosquitto.conf` → `allow_anonymous false` + TLS) and rerun the MQTT verification.
 - Use the **Policy** module to draft an "OT segmentation and zoning policy" and link it to NIS2 Art. 21(2) (b) and (e).
 - Run `nmap --script modbus-discover -p 5020 <VM2-IP>` to enumerate the slave IDs (1, 2, 3) with zero prior knowledge.
-

Checklist

- ☐ Meter admin registered as a product
- ☐ Nmap scan run; MQTT + Modbus + HTTP findings filed
- ☐ Anonymous MQTT subscribe verified
- ☐ Modbus unauth register read verified
- ☐ ZAP scan run; default creds + file-upload findings filed
- ☐ Semgrep scan run; hardcoded-secret finding filed
- ☐ OSV scan run; at least one dep advisory filed
- ☐ NIS2 assessment answered against the 10 questions in Step 7
- ☐ NIS2 readiness PDF generated